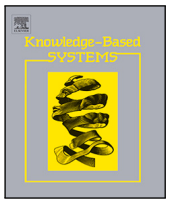
Knowledge-Based Systems 307 (2025) 112731



Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys





Can question-texts improve the recognition of handwritten mathematical expressions in respondents' solutions?

Ting Zhang *, Xinxin Jin, Xiaoyang Ma, Xinzi Peng, Yiyang Zhao, Jinzheng Liu, Xinguo Yu

Faculty of Artificial Intelligence in Education, Central China Normal University, Wuhan, China

ARTICLE INFO

Keywords:
Handwritten mathematical expressions
Handwriting recognition
Attention mechanism

ABSTRACT

The accurate recognition of respondents' handwritten solutions is important for implementing intelligent diagnosis and tutoring. This task is significantly challenging because of scribbled and irregular writing, especially when handling primary or secondary students whose handwriting has not yet been fully developed. Recognition becomes difficult in such cases even for humans relying only on the visual signals of handwritten content without any context. However, despite decades of work on handwriting recognition, few studies have explored the idea of utilizing external information (question priors) to improve the accuracy. Based on the correlation between questions and solutions, this study aims to explore whether question-texts can improve the recognition of handwritten mathematical expressions (HMEs) in respondents' solutions. Based on the encoder–decoder framework, which is the mainstream method for HME recognition, we propose two models for fusing question-text signals and handwriting-vision signals at the encoder and decoder stages, respectively. The first, called encoder-fusion, adopts a static query to implement the interaction between two modalities at the encoder phase, and to better catch and interpret the interaction, a fusing method based on a dynamic query at the decoder stage, called decoder-attend is proposed. These two models were evaluated on a self-collected dataset comprising approximately 7k samples and achieved accuracies of 62.61% and 64.20%, respectively, at the expression level. The experimental results demonstrated that both models outperformed the baseline model, which utilized only visual information. The encoder fusion achieved results similar to those of other state-of-the-art methods.

1. Introduction

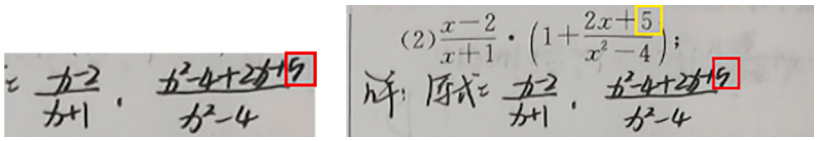
Collecting and perceiving student learning data are key issues that must be addressed in personalized learning. Learning data are the source through which the learning status of students can be diagnosed and tracked [1–4]. As paper-and-pencil exercises or tests are the main evaluation tools in daily teaching activities, handwritten responses to questions that reflect the fine-grained cognitive states of respondents are important parts of learning data. Recognizing and reading them accurately and rapidly is a problem encountered in realizing large-scale personalized learning. This study aimed to develop a method for recognizing the handwritten solutions of respondents as a preliminary technique to support intelligent diagnosis.

Mathematical expressions (MEs) are widely used in science, technology, engineering, and mathematics education because they represent knowledge intuitively and concisely, which is indispensable in the process of knowledge transmission. Naturally, they appear frequently in math or related assignments (both questions and responses). In most cases, MEs carry the main information regarding the solutions of respondents. Furthermore, from the perspective of recognition challenges, MEs have complex two-dimensional structures that are representative research objects in the document analysis and recognition field. From the above discussion, we focus on the problem of handwritten ME (HME) recognition in the solutions of the respondents.

HME recognition in general scenarios is challenging because of the following characteristics: (1) variable writing styles, (2) large sets of symbols, and (3) complex two-dimensional structures. Recognizing HMEs in the question-answering (QA) scenario is more difficult than in the general scenario owing to the variable, scribbling, and irregular writing style of students, especially junior students. This might be because their handwriting has not yet been fully developed at that stage. Even for humans, interpreting recognition ambiguities caused by scribbled writing is a challenge. However, there is a strong correlation between the questions and responses that can be exploited in this task. Symbols appearing in the questions are usually present in the responses

* Corresponding author.
E-mail addresses: ting.zhang@ccnu.edu.cn (T. Zhang), xinxinjin@mails.ccnu.edu.cn (X. Jin), xm779@uowmail.edu.au (X. Ma), pxz0220@mails.ccnu.edu.cn (X. Peng), zhaoyiyang@mails.ccnu.edu.cn (Y. Zhao), liujinzheng@mails.ccnu.edu.cn (J. Liu), xgyu@ccnu.edu.cn (X. Yu).

https://doi.org/10.1016/j.knosys.2024.112731
Received 1 May 2024; Received in revised form 30 October 2024; Accepted 11 November 2024
Available online 20 November 2024
0950-7051/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.



(a) (b)

Fig. 1. Illustration of handwritten math expressions. (a) without the question; (b) with the question. Red boxes highlight the recognition difficulties. Yellow box denotes the corresponding question context that can be exploited.

as well. Generally, people refer to question texts for help by combining visual features and question contexts to obtain interpretations. For example, as Fig. 1(a) shows, it is difficult to identify the handwritten mathematical symbols marked in the red box without question contexts; however, on considering question contexts (Fig. 1(b)), the handwritten mathematical symbol, previously unrecognizable, can now be easily identified as '5'.

Inspired by this, we explored the problem of improving the recognition of HMEs in the solutions of respondents prompted by questions.

After decades of work on handwriting recognition, significant progress has been made in terms of technology and accuracy. However, most studies have considered only the information of handwritten content itself, neglecting the importance of external information such as priors. However, these methods cannot solve the aforementioned problems encountered in present-day scenarios. Currently, mainstream methods for HME recognition are based on an encoder–decoder framework, which has been extensively developed in diverse fields such as machine translation [5], speech recognition [6], and image captioning [7]. These encoder–decoder based methods [8–11] regard HME recognition as an image-to-sequence task, taking a handwritten expression image as the input and generating the corresponding LaTeXmarkup sequence directly. Although some outstanding methods have been proposed and achieved good performance, they did not consider the ways to exploit question contexts in their methods. To address these limitations, we propose two models for introducing question context into the HME recognition task, both of which were developed from the encoder–decoder framework. (1) Encoder-fusion method. This model first extracts the visual features of the HME and textual features of the corresponding question. Subsequently, a novel multi-modal feature fusion module with a gate mechanism is proposed to fuse visual and textual features while being aware of their relevant mutual information. Finally, a fused representation is used to generate the predicted output. Overall, this approach adopted a static query to implement the interaction between the two modalities in the encoder phase. (2) Decoder-attend method. We propose a questionable attention-able decoder that introduces a question-textual attention mechanism in addition to a coverage visual attention mechanism. In each decoding step, the decoder captures both the question-textual features and visual contexts to enhance the recognition accuracy. Overall, this method uses a dynamic query to implement the interaction between two modalities in the decoder stage.

The contributions of this study are as follows:

- A new dataset QA-HMEs is proposed with $\approx 7k$ samples available in https://github.com/tingzhang-ccnu/QA-HMEs/tree/master. Compared with the existing benchmark datasets, our dataset includes HME images and corresponding question-text information.
- To the best of our knowledge, this is the first work on revealing the correlation between questions and responses and exploiting question context information in HME recognition. For this, two independent methods were proposed in this paper, the encoder-fusion method and the decoder-attend method. The experimental results demonstrate that both models outperformed the baseline model, where only visual signals are utilized.
- Unlike the principles of complementarity and consistency followed by current multi-modal fusion research, the problem explored in this paper pertains to the fusion of two sources of information that are not fully consistent, as questions and responses depict different objects. Therefore, this work could also boost the development of multi-modal fusion.

The remainder of this paper is organized as follows: Section 2 reviews related work; a self-collected QA-HMEs dataset is introduced in Section 3; Section 4 introduces the proposed methods in detail; experiments and results are presented in Section 5; and conclusions are presented in Section 6.

2. Related works

We review related works in two directions: HME recognition and multi-modal fusion, both of which are closely related to our work.

2.1. HME recognition

Research on HME recognition dates back to the 1960s. In the early stages, the proposed methods for HME recognition [12–16] generally followed the pipeline of symbol segmentation, symbol recognition, and structure analysis, which produced a set of symbol hypotheses first, and then, using the grammar parsing technique, generated a valid mathematical expression interpretation. This line of methods not only suffers from the error accumulation problem, but also requires manual design of numerous grammar productions.

In recent years, with the rapid development of deep learning techniques, the encoder–decoder framework has been widely explored in HME recognition. They regarded HME recognition as an image-to-sequence task that directly transcribes a handwritten expression image into the corresponding LaTeXmarkup sequence. Benefiting from their end-to-end trainable property, encoder–decoder-based methods significantly outperform traditional methods. Deng et al. [17] used a convolutional neural network and recurrent neural network (RNN) as the encoders for feature extraction, and an RNN with a coarse-to-fine attention mechanism as the decoder to generate the LaTeXsequence step-by-step. Zhang et al. [8] proposed the Watch, Attend, and Parse (WAP) model. WAP model applies a fully convolutional network (FCN) as the encoder and utilizes coverage attention [18], which is the sum of all past attention weights, to alleviate the lack of coverage problem. Subsequently, based on the WAP model, Zhang et al. [19] presented further improvements by replacing the FCN with the densely connected convolutional network (DenseNet) [20] as the encoder and proposed a multi-scale attention mechanism to consider variable scales of symbols in mathematical formulas. Zhao et al. [10] applied a transformer [21] instead of RNN as the decoder, used positional encoding to mitigate the lack of coverage problem, and enhanced the decoding capabilities with bi-directional language modeling. Bian et al. [11] proposed an attention-aggregation-based bidirectional mutual learning network comprising a shared encoder and two inverse decoders to learn complementary context information. Li et al. [22] designed a counting module that can be plugged into an attention-based encoder–decoder framework, taking the counting results of each symbol class as supplementary information to enhance the recognition accuracy. Fu

et al. [23] proposed a symbol location-aware network that first detects symbols using a relationship-level counting method and then decodes the sorted symbol sequence using a global context-aware decoder. Liu et al. [24] proposed a semantic-aware method that enhances the understanding of a model for symbolic relationships by jointly optimizing mathematical formula recognition tasks and semantic interaction learning between symbols. Wen et al. [25] proposed a DGNet based on deformable convolution and global context attention, which replaced the convolution module of DenseNet with a deformable module during the encoding stage, and used global context attention in the decoding stage to alleviate the long-distance dependency problem of complex formulas. Guan et al. [26] proposed a position-forest transformer for HME recognition. The model optimized both expression and position recognition to facilitate position-aware feature learning for symbols in mathematical expressions. In addition, some methods have applied the tree decoder to explicitly model the parent–child relationship of mathematical expression trees. Zhang et al. [9] proposed a tree-structure decoder to generate a subtree sequence containing parent–child relationships. Yuan et al. [27] proposed a syntax-aware network that integrates syntax information into an encoder–decoder model, alleviating the tree-structure ambiguity by dividing the syntax tree into different components.

In general, these deep-learning-based methods are either devoted to optimizing the encoder or decoder, or introducing additional information, such as counting information and grammar rules, to promote the recognition accuracy. However, the information they introduce is limited to the handwritten content itself, neglecting external information such as question contexts. Therefore, they cannot solve the problem of recognition ambiguity caused by scribbling in QA scenarios. To fill this gap, we propose a solution that considers both handwritten content and external information (question text).

2.2. Multi-modal fusion method

Existing multi-modal fusion methods can be roughly divided into three classes: early, intermediate, and late fusion. Early fusion [28,29], also known as feature-level fusion, directly integrates the features extracted from each modality into a single high-dimensional feature vector. Late fusion [30,31], also known as decision-level fusion, first makes decisions based on each modality and then combines these decisions by applying different rules, such as averaging [32] and majority voting [33]. However, neither the early nor late fusion methods can fully exploit intra-modal and inter-modal interactions.

Intermediate fusion, which is more flexible, refers to the fusion process conducted in the intermediate layers of a deep model. You et al. [34] integrated a tree-structured long short term memory (LSTM) network with a visual attention mechanism to capture the correlation between image and text. In [35], a mixed-fusion framework that integrates intermediate and late fusion was proposed for multi-modal sentiment analysis. Huang et al. [36] associated linguistic features with correlated visual features by using bilinear pooling [37]. In [38], a deep co-attention model comprising multiple co-attention layers was proposed to fuse multi-modality features for fake news detection. Yang et al. [39] integrated linguistic features into visual features using a hierarchy of visual transformer layers. TransFuser [40] was proposed to fuse images and keyword information into a final attention-based embedding vector. Wang et al. [41] proposed a fine-grained multi-modal fusion network that combined multiple visual features of an image with text features. Chen et al. [42] proposed a multi-level fusion strategy equipped with a coarse-grained prefix-guided interaction and fine-grained correlation-aware module to integrate visual and text representations for multi-modal knowledge graph completion.

In this study, multi-modal fusion is used because we need to consider both the visual features of HMEs and the question-textual features to make a final decision. One difference was that the two modalities in our task did not follow the principles of consistency or complementarity, which are common in traditional multi-modal fusion tasks. The questions and responses depict different objects.

Table 1
The number of expression–question pairs under different segments of R_n.

R_n	Number
≤ 0.2	154
0.2–0.4	1373
0.4–0.6	2054
0.6–0.8	1978
≥ 0.8	1332

3. QA-HMEs dataset

In this study, we explored whether question texts can improve the recognition of HMEs in the solutions of respondents. To the best of our knowledge, no dataset comprising QA pairs is available yet. Existing public datasets CROHME [43] and HME100K [27] contain only online or offline handwritten mathematical expressions. Therefore, we built a new dataset called QA-HMEs. Test papers and assignments of middle school students were collected and processed to meet our requirements, with private information masked. Considering Fig. 2 as an example, the student answered the question in three steps that were segmented manually. These three HMEs shared the same question: (1 + \frac { 4 } { a − 2 }) \div \frac { a ^ { 2 } + 4 a + 4 } { 3 a + 6 }. In total, a dataset comprising 6902 expression–question pairs was built. Considering both questions and answers, there were 109 symbol classes, including “sos”, “eos”, and “#”. The “sos” and “eos” were added to the LaTeXsequences as the start and end symbols, respectively, during training; the “#” indicates the blacked-out symbol. We split the dataset into training, validation, and test sets at a ratio of 8:1:1, containing 5521, 691, and 690 samples, respectively. Each sample comprised an expression image and corresponding question text.

3.1. Statistical analysis of the correlation between questions and responses

As previously mentioned, there is a correlation between questions and responses. The symbols appearing in the questions are generally present in responses. Based on whether the question symbols appear in the response expressions and how many question symbols appear in the response expressions, we define two metrics to statistically analyze the correlation between the questions and response expressions:

$$R_e = \frac{N_q}{N} \tag{1}$$

$$R_n = \frac{Q_n}{Q} \tag{2}$$

where R_e represents the ratio of response expressions containing question symbols (no. ≥ 1) for all expressions in the QA–HMEs dataset, N_q is the number of response expressions containing question symbols, and N is the total number of response expressions. R_n is defined to represent the ratio of question symbols appearing in the response expressions to all question symbols for each expression–question pair, where Q_n is the number of question symbols appearing in the response expressions and Q is the total number of symbols in the question.

According to the statistical results, there are 6891 expressions containing question symbols among all samples. Specifically, the ratio R_e is $6891/6902 \approx 99.84\%$. We split the R_n metric into several segments with ≤ 0.2, 0.2–0.4, 0.4–0.6, 0.6–0.8, and ≥ 0.8. The number of expression–question pairs for each segment is listed in Table 1.

4. Methods

Based on the correlation between questions and answers, we introduced the question-text information to improve the recognition accuracy of HMEs in the solutions of the respondents. To achieve this, two methods were proposed in this study. Both methods employ the cross attention mechanism to implement the interaction between two modalities. Method I adopts a static query, while Method II uses a

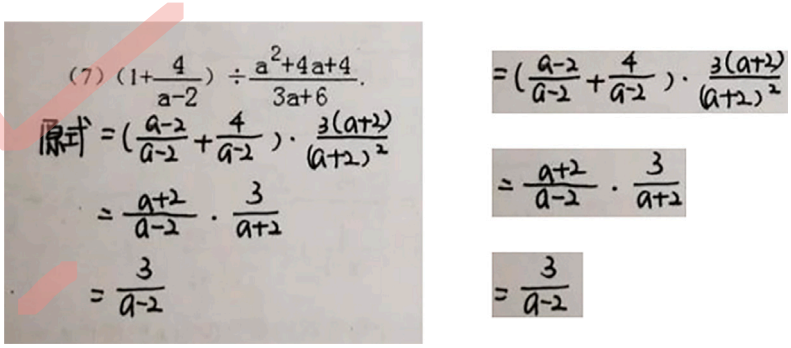


Fig. 2. Illustration of collected samples.

dynamic query. (1) Encoder-fusion: This method adopts at the encoder stage a multi-modal feature fusion module with a gate mechanism to fuse the visual features of the HME and textual features of the related question, capturing their relevant mutual information. (2) Decoder-attend: This method is an improvement over conventional decoders. In addition to the equipped visual attention mechanism, we propose a question–attention module. In each decoding step, the module captures the question features related to the visual context. Both methods use DenseNet [20] as the backbone network for extracting visual features. The dense connections between layers benefit the extraction and fusion of multi-scale features, which helps consider the variable scales of symbols in HMEs. In the following subsections, we elaborate on the two methods.

4.1. Problem formulation

The problem of recognizing HMEs in the solutions of the respondents is clearly defined as follows: given the question text Q and one HME image X as a part of the response solution, the task is to find the target markup Y that represents the expression image X unambiguously. The recognition system learns to generate Y by maximizing the conditional probability $P(Y|X,Q)$.

4.2. Method I : Encoder-fusion

In this subsection, we introduce the incorporation of question-text information into the encoding stage. The framework of the proposed method is illustrated in Fig. 3, from which it can be seen that three modules are involved: the feature extraction, feature fusion, and decoder. First, an HME image and its corresponding question sequence are fed into two independent extractors to generate visual and question-textual features. The extracted features are then fused through the feature fusion module, in which the scaled dot-product attention mechanism is used to acquire the image-related question feature, and a gating mechanism is used to control the fusion of the attended feature with the visual feature. Finally, the fused feature was used as the input of the gated recurrent unit (GRU)-based decoder to generate the predicted sequence step-by-step.

4.2.1. Feature extraction

Given an input pair of an HME image and its corresponding question text, we utilized two distinct models to extract visual features and question-textual features.

Visual feature. Following [19], we adopt DenseNet [20] to extract visual features from the HME image. The output is a three-dimensional feature map $F \in \mathbb{R}^{H\times W\times C}$, where H, W and C denote the height, width, and channel, respectively.

Question-textual feature. The correlation between questions Q and HMEs is simplified as the correlation between math symbols or math expressions Q_m contained in questions and HMEs in this work. These symbols or expressions are represented as LaTeXsequences to be handled by the textual feature extractor. Thus, in addition to visible mathematical symbols, invisible and auxiliary tokens also exist in question texts, such as {, }, ∧, _. An embedding matrix $W_e \in \mathbb{R}^{E\times S}$ is employed to embed the simplified question text Q_m into a high-dimensional vector $L \in \mathbb{R}^{E\times T}$ as a question-textual feature.

$$L = W_e Q_m \tag{3}$$

where E denotes the token embedding size, S denotes the number of symbol classes, and T denotes the length of the simplified question text Q_m.

4.2.2. Feature fusion module

To model the interaction between HME visual information and question-textual information, we proposed a feature fusion module based on the scaled dot product attention mechanism [21]. In contrast to traditional multi-modal fusion tasks, only a part (not all) of the question texts is useful for our recognition task in most cases. Thus, a scaled dot product attention mechanism was adopted to calculate the importance of different tokens based on visual features. Specifically, with the visual feature F as the query and question-textual feature L as the key and value, the attended question-textual feature Z can be computed using the following equations:

$$Q = W_q(F) \tag{4}$$

$$K = W_k(L) \tag{5}$$

$$V = W_v(L) \tag{6}$$

$$Z = Attention(Q,K,V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \tag{7}$$

where W_q, W_k, W_v are the projection functions. The key and value projections W_k and W_v are implemented with a 1×1 convolution, whereas the query projection W_q is implemented with a 1×1 convolution, followed by instance normalization. Fig. 4 illustrates the computation of the attended question features.

As mentioned previously, the question text works as a hint for the recognition task when required. If a visual feature is sufficiently clear to support the recognition process, a hint is no longer required. Therefore, to prevent unnecessary negative effects of textual features on visual features, a gating mechanism is used to control the fusion of visual and attended textual features, as shown in Eq. (8):

$$F_f = g(Z)Z + F \tag{8}$$

where $g(\cdot)$ represents the gating mechanism, which is a two-layer perceptron, with the first layer being a linear layer followed by a rectified linear unit nonlinearity activation, and the second layer is a linear layer followed by a Tanh function. F_f represents the fused features. Fig. 5 illustrates the operation of this gating mechanism.

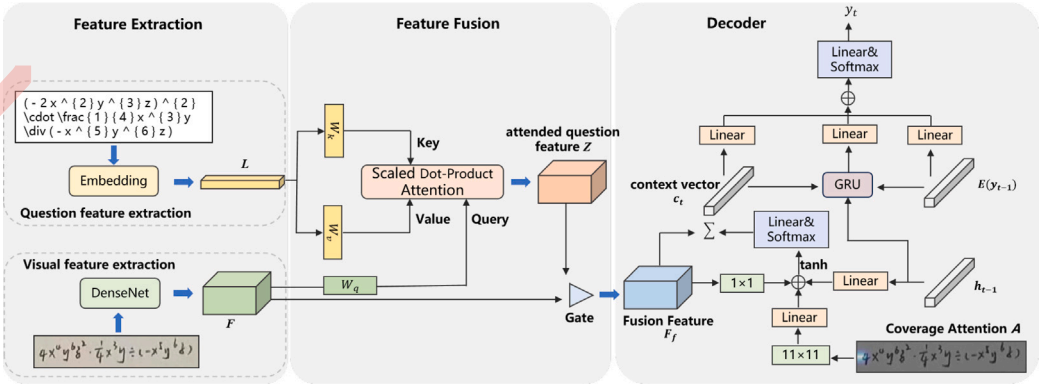


Fig. 3. The overall structure of the method of fusing features at the encoding stage (encoder-fusion). ⊕ indicates the element-wise sum.

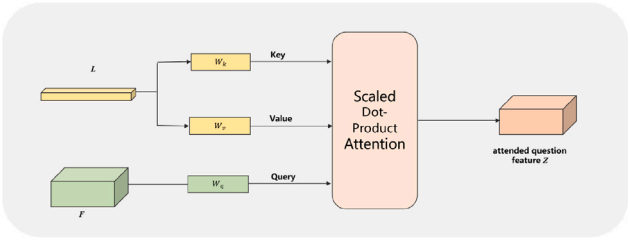


Fig. 4. Illustration of the interaction between the HME visual information and the question-textual information.

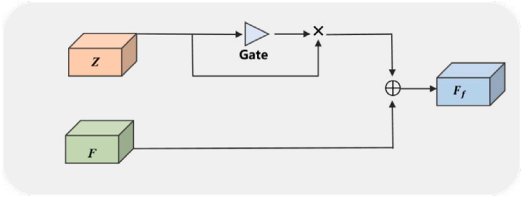


Fig. 5. Illustration of the gating mechanism employed in the fusing process. × indicates the multiplication and ⊕ indicates the element-wise sum.

4.2.3. Decoder

Both LSTM and GRU are variants of RNN and can alleviate the vanishing and exploding gradient problems. Compared to LSTM, GRU has a similar performance but a lower computational cost. Therefore, as in [8,22], we have used GRU in our experiments. A GRU-based decoder with an attention mechanism has been used to decode the fused features to produce the LaTeX sequence. When decoding at step t, the previous hidden state h_{t-1}, fused feature F_f and coverage attention A are utilized to compute the current attention weight α_t and context vector c_t:

$$e_t = v_a^T \tanh(W_h h_{t-1} + U_f F_f + W_a A) \tag{9}$$

$$\alpha_{t,ij} = \frac{exp(e_{t,ij})}{\sum_{p=1}^{H}\sum_{q=1}^{W} exp(e_{t,pq})} \tag{10}$$

$$c_t = \sum_{i=1}^{H\times W} \alpha_{t,i} F_{f_i} \tag{11}$$

where v_a, W_h, W_a are the trainable weights, U_f is a 1×1 convolution operation, and A is the sum of all past attention weights. Then, we feed the embedding of the decoded symbol y_{t-1} and context vector c_t into a GRU cell to obtain the current hidden state h_t.

$$h_t = GRU(E(y_{t-1}), c_t, h_{t-1}) \tag{12}$$

Finally, the decoded symbol probability is computed as follows:

$$p(y_t) = softmax(w_o^T(W_e E(y_{t-1}) + W_{h'} h_t + W_c c_t)) \tag{13}$$

where w_o, W_e, $W_{h'}$, W_c are the trainable weights.

4.3. Method II : Decoder-attend

In Method I, the global visual feature has been adopted as a static query to implement the interaction between the two modalities in the encoder phase. Because the encoder–decoder framework predicts the labels step-by-step, it is interesting to explore how the question text information works at different decoding steps. Therefore, to better catch and interpret the interaction between two modalities, we propose a fusion method based on a dynamic query at the decoder stage.

As with Method I, we first utilize two independent models to extract the visual and question-textual features, respectively, generating F and L. Subsequently, both F and L are fed into the proposed attention-able decoder to output a sequence of labels. We do not repeat the feature extraction process described in Section 4.2.1 and focus only on the introduction of question attention-able decoder.

4.3.1. Question attention-able decoder

Intuitively, people look at the HME images first and then read the question information in the case of scribbled or ambiguous writing. Inspired by this observation, a novel attention-able decoder was developed that attends to the visual context first and then the question context. The structure of the proposed decoder is shown in Fig. 6.

Given the extracted visual features F and L, when decoding at step t, the decoder first attends to the visual context c_t^v, which can be computed as follows:

$$e_t^v = v^T \tanh(W_h^v h_{t-1} + U_f^v F + W_a^v A) \tag{14}$$

$$\alpha_{t,ij}^v = \frac{exp(e_{t,ij}^v)}{\sum_{p=1}^{H}\sum_{q=1}^{W} exp(e_{t,pq}^v)} \tag{15}$$

$$c_t^v = \sum_{i=1}^{H\times W} \alpha_{t,i}^v F_i \tag{16}$$

where v, W_h^v, W_a^v are trainable weights, U_f^v is a 1×1 convolution operation, and A is the sum of all past attention weights. The visual attention module works similarly to that in Method I.

Next, a question attention module is developed to obtain the question context. The visual context vector c_t^v denotes the dominant visual features attended to by the decoder when making decisions in step t. We have used it as a query for the question attention module. By measuring the similarity between the visual context vector c_t^v and question feature L, the dominant question features are emphasized and stored as c_t^l. The calculation process is as follows:

$$\alpha_t^l = f_{atten}(L, c_t^v) \tag{17}$$

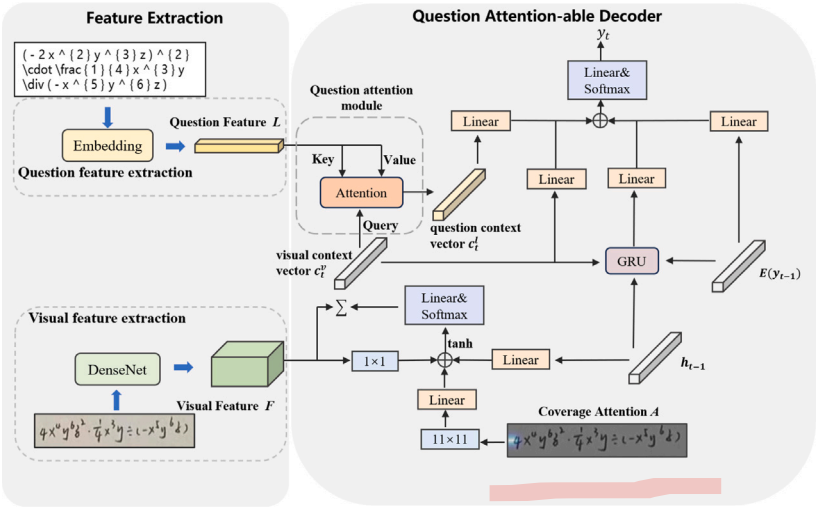


Fig. 6. The overall structure of the method of fusing features at the decoding stage (Decoder-Attend).

$$c_t^l = \sum_i \alpha_{t,i}^l L_i \tag{18}$$

where α_t^l is the weight of the question attention weight. The f_{atten} function is designed as follows:

$$e_t^l = v_l^T \tanh(W_c^v c_t^v + W_l L) \tag{19}$$

$$\alpha_{t,i}^l = \frac{exp(e_{t,i}^l)}{\sum_k exp(e_{t,k}^l)} \tag{20}$$

Subsequently, the embedding of the decoded label y_{t-1} and visual context vector c_t^v are fed into a GRU cell to obtain the current hidden state h_t. Finally, the attended visual context vector c_t^v, attended question context vector c_t^l, hidden state h_t, and embedding $E(y_{t-1})$ are combined to make predictions.

$$p(y_t) = softmax(w_o^T(W_e^l E(y_{t-1}) + W_h^l h_t + W_{cv} c_t^v + W_{cl} c_t^l)) \tag{21}$$

where w_o, W_e^l, W_h^l, W_{cv}, W_{cl} are the trainable weights.

4.4. Loss function

Both methods I and II are trained end-to-end with the objective of minimizing the distance between the ground truths and predictions. In this study, the cross-entropy classification loss function was used to measure the loss between them.

$$O = -\sum_{t=1}^{T} g_t \log p(y_t) \tag{22}$$

where g_t denotes the ground truth at step t.

5. Experiments

5.1. Datasets and metrics

Datasets. We used the proposed QA-HMEs dataset to train and evaluate our models. The QA-HMEs training set comprises 5521 expression–question pairs, while the validation and test sets contain 691 and 690 pairs, respectively. Totally, there are 109 symbol classes.

Metrics. The expression recognition rate (ExpRate) is the widely used recognition protocol for mathematical expression recognition, defined as the percentage of correctly predicted mathematical expressions. Moreover, ≤ 1, ≤ 2, and ≤ 3 were used, indicating that the expression recognition rate is tolerable for at most one to three symbol-level errors.

Table 2
Comparison with previous works (in %) on QA-HMEs.

Datasets	Methods	ExpRate	≤ 1	≤ 2	≤ 3
QA-HMEs	WAP [8]	45.94	64.64	73.91	79.57
	BTTR [10]	65.36	81.59	88.99	91.16
	ABM [11]	64.06	80.72	88.55	91.16
	CAN [22]	64.64	81.01	87.83	91.01
	SAN [27]	63.77	81.74	88.26	90.87
	CoMER [46]	66.96	83.91	88.84	91.59
QA-HMEs	DWAP (baseline)	62.90	78.12	87.25	90.43
	Encoder-Fusion (ours)	62.61	80.58	88.26	90.87
	Decoder-Attend (ours)	64.20	80.29	87.10	90.87

5.2. Implementation details

We implemented the proposed methods using PyTorch and all experiments were conducted using a single Nvidia GeForce RTX 2080 Ti with 11G memory. For the visual feature-extractor parameters, DenseNet was stacked with three dense blocks and two transition layers. The dense block comprises 16 dense layers with 1×1 convolution and 3×3 convolution. The transition layer between the two dense blocks comprised 1×1 convolution and 2×2 average pooling. The final dimensions of the extracted visual features were 684. For the embedding matrix used to extract the question textual features, the embedding dimensions were set to 256. For the decoder, the hidden state size of the GRU was set to 256 and the dimensions of the word embedding were 256. During the training process, we employ the Adadelta algorithm [44] for optimization. The learning rate starts from 0, monotonously increases to 1 at the end of the first epoch, and decays to zero following cosine schedules [45]. We set the batch size to four and the total number of training epochs to 200.

5.3. Comparison with previous works

To compare our methods with those of previous studies, we selected several recent and dominant methods in HME recognition and evaluated them on QA-HMEs. To ensure the fairness of the performance comparison, none of the experiments used data augmentation.

As shown in Table 2, we can observe the following: (1) Compared with the baseline method DWAP (WAP [8] with DenseNet as the encoder), both the proposed methods exhibit better performance. The encoder-fusion method exhibits the same performance as that of the baseline model on the ExpRate, but significantly outperforms the baseline on ≤ 1 (2.46% ↑) and ≤ 2 (1.01% ↑). Moreover, the decoder–attention method outperforms the baseline by margins of 1.30% ↑

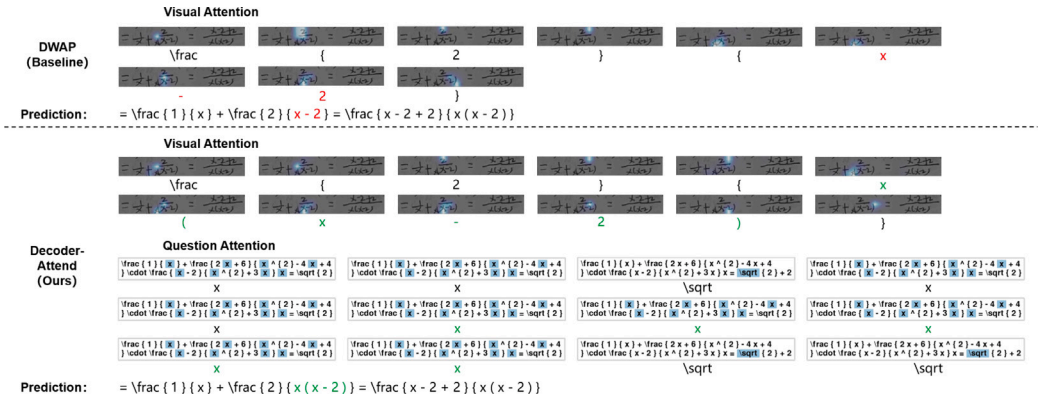


Fig. 7. Attention visualization of one sample processed by DWAP and decoder-attend.

Table 3
Comparison results (in %) in the hyper-parameter ablation experiment. Model: DWAP; dataset: QA-HMEs.

Depth	ExpRate	≤ 1	≤ 2	≤ 3
4	49.57	67.25	78.41	83.33
8	58.99	77.97	86.81	89.57
16	62.90	78.12	87.25	90.43
24	63.77	78.84	87.68	91.01
32	63.48	78.99	86.96	90.14

Table 4
Comparison results (in %) in the component ablation experiment.

Method	ExpRate	≤ 1	≤ 2	≤ 3
Baseline	62.90	78.12	87.25	90.43
+ Fusion Module	57.83	73.48	82.75	86.38
+ Fusion Module + Gate	62.61	80.58	88.26	90.87
+ Question Attention Module	64.20	80.29	87.10	90.87

and 2.17% ↑ on the ExpRate and ≤ 1, respectively. These results demonstrate the usefulness of introducing question texts in this task, as well as the effectiveness of the proposed feature fusion module and question attention-able decoder.

(2) Compared to previous methods, decoder-attend achieved similar results to that of other SOTA methods such as ABM [11] and CAN [22]. However, there remains a small gap between the encoder fusion and SOTA methods. Encoder-fusion did not show a significant improvement in terms of the ExpRate compared with the baseline. There are two possible reasons for this. First, although there is a correlation between questions and HMEs, the two modalities do not describe the same object. Certain symbols in the questions may not have appeared in the responses. This redundant information may have negative effects on HME recognition (detailed analysis in Section 5.4). Second, because of the length of mathematical expressions, HME recognition is a step-by-step process. This implies that the fusion of the two modes is not accomplished in a single stroke, it is dynamic and multi-step. However, the encoder-fusion method fuses at one strokes, which is unreasonable. The decoder-attend method adopts a dynamic multi-step fusion mechanism. An improvement in ExpRate was observed compared to the baseline.

5.4. Ablation study

Hyper-parameter. In order to set the appropriate depth of the dense block in DenseNet on QA-HMEs, we conducted a hyper-parameter ablation experiment. As can be seen in Table 3, compared to models with a depth of four or eight, the performance of a depth of 16 was significantly improved. When the depth was further increased to 24 and 32, the performance was only slightly improved; however, the computing resources were significantly increased. As a tradeoff, we set the depth of the dense block to 16 in our experiments.

Component analysis. In order to verify the effectiveness of the proposed fusion module, gating mechanism, and question attention module, we conducted the component ablation experiment. The results are presented in Table 4.

- Baseline: The baseline model is a traditional encoder–decoder architecture (DWAP).
- + Fusion Module: The attended question-textual feature via the scaled dot-product attention mechanism is directly integrated with the visual feature without the gating strategy.
- + Fusion Module + Gate: Instead of combining the attended question-textual feature and visual feature directly, a gating mechanism is adopted to control the flow of the attended question-textual feature first.
- + Question Attention Module: A question attention module is introduced into the traditional decoder of the baseline with other parts being the same.

From the ablation experiment results, we can see the following: + **Fusion Module** incurs negative influences (decreases on all metrics), which is reasonable, as explained previously, questions and HMEs in answers do not describe the same object. Certain symbols in the questions may not have appeared in the responses. This redundant information may have negative effects on HME recognition. Thus, the mechanisms must be proposed to control the flow of question-textual features. When we further introduce the gating strategy, the accuracy increases to the same level as that of the baseline in terms of the ExpRate and significantly outperforms the baseline on ≤ 1 and ≤ 2. Thus, the proposed gating mechanism can effectively control the flow of question-textual features during the fusion process, thereby avoiding a negative influence on visual features. When we kept the baseline encoder unchanged and introduced the question-attention module into the traditional decoder, significant improvements in almost all metrics were observed, verifying the effectiveness of the proposed question-attention module.

5.5. Attention visualization

To further reveal the internal working mechanism of the decoder-attend method, we have visualized the attention map of certain decoding steps for each sample. Unlike the traditional decoder, which only attends to the visual feature map, the question attention-able decoder is able to attend to both the visual and question features. Therefore, in addition to the visual attention map, we visualized the attention distribution for the question. For each decoding step, we highlight the symbols in the question that received the highest attention. As shown in Fig. 7, the ground truth is $= \frac{1}{x} + \frac{2}{x(x-2)} = \frac{x-2+2}{x(x-2)}$, while the

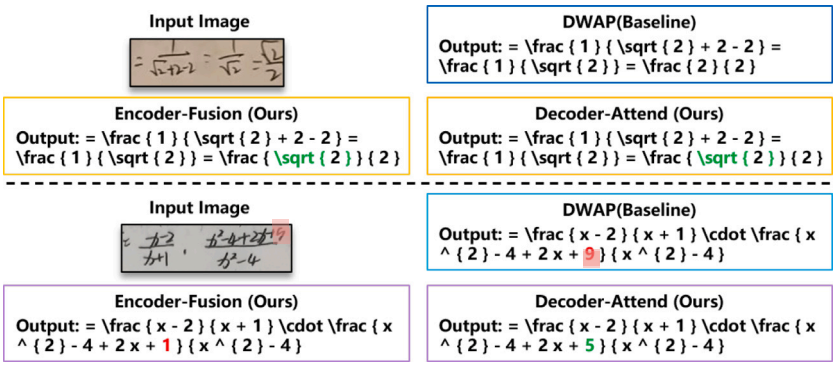


Fig. 8. Recognition results of DWAP and the proposed method.

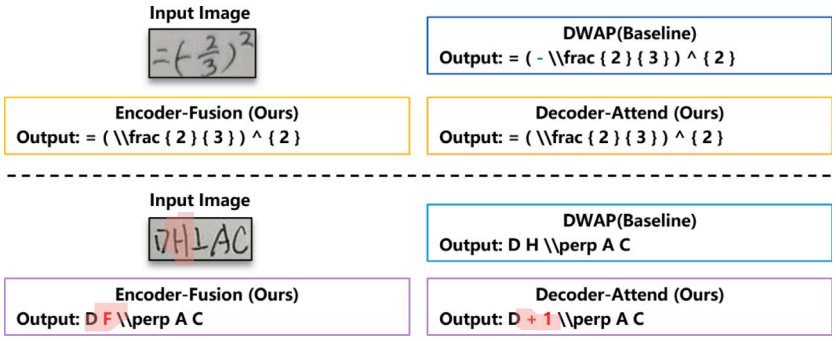


Fig. 9. Recognition results of DWAP and the proposed method.

prediction of DWAP is $= \frac{1}{x} + \frac{2}{x-2} = \frac{x-2+2}{x(x-2)}$. DWAP misses the second x because the corresponding visual representation is unclear. In contrast, decoder-attend outputs the correct predictions. As can be seen from the question-attention visualization, when decoding the three steps of "x (x", the question-attention module attends more on the symbol "x" in the question-text. Decoder-attend synergizes visual attention and question attention to make a correct prediction. Furthermore, as a synergistic effect, visual attention is more accurate compared with DWAP.

5.6. Performance on variable sequence length

To explore whether our methods are more friendly to expressions of a certain length, we have split the test dataset into several subsets according to the length of the expressions in the LaTeXsequences to conduct a deep analysis. Typically, the longer the expression, the more difficult it is to recognize. From Table 5, it can be seen that: (1) compared with the baseline model, encoder fusion shows an obvious drop in ExpRate when considering expressions with the sequence length ≤ 10, but exhibits advantages for sequence length > 10. (2) Owing to the dynamic and multi-step fusion mechanism, decoder-attend presents superior performance to the baseline method when the sequence length is > 10, whereas it remains at a similar level for expressions ≤ 10.

Overall, both encoder-fusion and decoder-attend exhibited superior results for expression > 10, but neither showed significant improvements in expression ≤ 10. We conducted a case study to identify the underlying reasons. Figs. 8 and 9 show some recognition samples for DWAP and our methods. For the second sample in Fig. 8, the baseline method interpreted as '5' as '9" because of ambiguous writing, whereas decoder-attend recognized it as '5' correctly when the question context was considered ('5' appears in the question). For the sample $DH \perp AC$ illustrated in Fig. 9, the baseline model recognized it correctly but encoder-fusion interpreted it as $DF \perp AC$ as 'F' appears in the question while 'H' does not. When considering simple and short expressions, visual features are sufficient to support the recognition process, and the introduction of question text in these cases has more negative than positive impacts. From these detailed results, we can see that the introduction of the question text in this task was a double-edged sword. Our main aim is to reduce this negative impact.

Table 5
ExpRate (in %) of variable sequence length.

Methods	Length			
	1–10	11–20	21–30	30+
BTTR [10]	76.68	70.82	49.09	44.00
ABM [11]	75.78	69.65	44.55	45.00
CAN [22]	75.78	69.65	47.27	46.00
SAN [27]	79.37	69.26	40.91	40.00
CoMER [46]	78.92	75.88	47.27	46.00
DWAP (baseline)	78.48	66.93	41.82	41.00
Encoder-Fusion (ours)	74.89	67.70	43.64	43.00
Decoder-Attend (ours)	78.92	68.48	47.27	44.00

6. Conclusion

In this study, efforts were devoted to improving the recognition of HMEs in the solutions from respondents by introducing a question context. Two methods were proposed to model the interactions between visual and textual signals. Encoder-fusion proposed the utilization of a static query to implement the interaction between two modalities at the encoder phase, whereas decoder-attend adopted a dynamic query that is more flexible at the decoder stage. The proposed method outperformed the baseline model, verifying the effectiveness of introducing question context. However, owing to limitations, the dataset collected in this study was relatively small. In the future, we will address this limitation by introducing large vision–language pre-trained models.

CRediT authorship contribution statement

Ting Zhang: Writing – review & editing, Supervision, Conceptualization. **Xinxin Jin:** Writing – original draft, Software, Methodology. **Xiaoyang Ma:** Software. **Xinzi Peng:** Validation, Data curation. **Yiyang Zhao:** Investigation, Data curation. **Jinzheng Liu:** Software. **Xinguo Yu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study was supported by the National Natural Science Foundation of China (Grant No. 62007014).

Data availability

I have shared the link to my data in the paper.

References

[1] C. Piech, J. Bassen, J. Huang, S. Ganguli, M. Sahami, L.J. Guibas, J. Sohl-Dickstein, Deep knowledge tracing, Adv. Neural Inf. Process. Syst. 28 (2015).

[2] F. Wang, Q. Liu, E. Chen, Z. Huang, Y. Chen, Y. Yin, Z. Huang, S. Wang, Neural cognitive diagnosis for intelligent education systems, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, 2020, pp. 6153–6161.

[3] Y. Yang, J. Shen, Y. Qu, Y. Liu, K. Wang, Y. Zhu, W. Zhang, Y. Yu, GIKT: A graph-based interaction model for knowledge tracing, in: Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD 2020, Ghent, Belgium, September 14–18, 2020, Proceedings, Part I, Springer, 2021, pp. 299–315.

[4] T. Qi, M. Ren, L. Guo, X. Li, J. Li, L. Zhang, ICD: A new interpretable cognitive diagnosis model for intelligent tutor systems, Expert Syst. Appl. 215 (2023) 119309.

[5] K. Cho, B. Van Merrienboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, Y. Bengio, Learning phrase representations using RNN encoder-decoder for statistical machine translation, Comput. Sci. (2014).

[6] K. Xu, J. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhutdinov, R. Zemel, Y. Bengio, Show, attend and tell: Neural image caption generation with visual attention, Comput. Sci. (2015) 2048–2057.

[7] D. Bahdanau, J. Chorowski, D. Serdyuk, P. Brakel, Y. Bengio, End-to-end attention-based large vocabulary speech recognition, in: 2016 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, 2016.

[8] J. Zhang, J. Du, S. Zhang, D. Liu, Y. Hu, J. Hu, S. Wei, L. Dai, Watch, attend and parse: An end-to-end neural network based approach to handwritten mathematical expression recognition, Pattern Recognit. (2017) 196–206.

[9] J. Zhang, J. Du, Y. Yang, Y.-Z. Song, S. Wei, L. Dai, A tree-structured decoder for image-to-markup generation, in: International Conference on Machine Learning, 2020.

[10] W. Zhao, L. Gao, Z. Yan, S. Peng, L. Du, Z. Zhang, Handwritten mathematical expression recognition with bidirectionally trained transformer, in: IEEE International Conference on Document Analysis and Recognition, 2021.

[11] X. Bian, B. Qin, X. Xin, J. Li, X. Su, Y. Wang, Handwritten mathematical expression recognition via attention aggregation based bi-directional mutual learning, in: AAAI Conference on Artificial Intelligence, 2021.

[12] S. Lavirotte, L. Pottier, Mathematical formula recognition using graph grammar, in: Document Recognition V, vol. 3305, SPIE, 1998, pp. 44–52.

[13] K.-F. Chan, D.-Y. Yeung, An efficient syntactic approach to structural analysis of on-line handwritten mathematical expressions, Pattern Recognit. 33 (3) (2000) 375–384.

[14] S. MacLean, G. Labahn, A new approach for recognizing handwritten mathematics using relational grammars and fuzzy sets, Int. J. Doc. Anal. Recognit. 16 (2013) 139–163.

[15] F. Alvaro, J.-A. Sánchez, J.-M. Benedí, Recognition of on-line handwritten mathematical expressions using 2D stochastic context-free grammars and hidden Markov models, Pattern Recognit. Lett. 35 (2014) 58–67.

[16] F. Álvaro, J.-A. Sánchez, J.-M. Benedí, An integrated grammar-based approach for mathematical expression recognition, Pattern Recognit. 51 (2016) 135–147.

[17] Y. Deng, A. Kanervisto, J. Ling, A.M. Rush, Image-to-markup generation with coarse-to-fine attention, in: International Conference on Machine Learning, PMLR, 2017, pp. 980–989.

[18] Z. Tu, Z. Lu, Y. Liu, X. Liu, H. Li, Modeling coverage for neural machine translation, 2016, arXiv preprint arXiv:1601.04811.

[19] J. Zhang, J. Du, L. Dai, Multi-scale attention with dense encoder for handwritten mathematical expression recognition, in: 2018 24th International Conference on Pattern Recognition, ICPR, IEEE, 2018, pp. 2245–2250.

[20] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 4700–4708.

[21] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, Adv. Neural Inf. Process. Syst. 30 (2017).

[22] B. Li, Y. Yuan, D. Liang, X. Liu, Z. Ji, J. Bai, W. Liu, X. Bai, When counting meets HMER: Counting-aware network for handwritten mathematical expression recognition, in: European Conference on Computer Vision, Springer, 2022, pp. 197–214.

[23] Y. Fu, W. Cai, M. Gao, A. Zhou, Symbol location-aware network for improving handwritten mathematical expression recognition, in: Proceedings of the 2023 ACM International Conference on Multimedia Retrieval, 2023.

[24] Z. Liu, Y. Yuan, Z. Ji, J. Bai, B. Xiang, Semantic graph representation learning for handwritten mathematical expression recognition, in: Proceedings of the 2023 International Conference on Document Analysis and Recognition, 2023.

[25] C. Wen, L. Yin, S. Liu, DGNet: A handwritten mathematical formula recognition network based on deformable convolution and global context attention, Mob. Netw. Appl. (2024).

[26] T. Guan, C. Lin, W. Shen, X. Yang, PosFormer: Recognizing complex handwritten mathematical expression with position forest transformer, in: Proceedings of the 2024 European Conference on Computer Vision, 2024.

[27] Y. Yuan, X. Liu, W. Dikubab, H. Liu, Z. Ji, Z. Wu, X. Bai, Syntax-aware network for handwritten mathematical expression recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 4553–4562.

[28] S. Poria, I. Chaturvedi, E. Cambria, A. Hussain, Convolutional MKL based multimodal emotion recognition and sentiment analysis, in: 2016 IEEE 16th International Conference on Data Mining, ICDM, IEEE, 2016, pp. 439–448.

[29] S. Park, H.S. Shim, M. Chatterjee, K. Sagae, L.-P. Morency, Multimodal analysis and prediction of persuasiveness in online social multimedia, ACM Trans. Interact. Intell. Syst. 6 (3) (2016) 1–25.

[30] M. Wang, D. Cao, L. Li, S. Li, R. Ji, Microblog sentiment analysis based on cross-media bag-of-words model, in: Proceedings of International Conference on Internet Multimedia Computing and Service, 2014, pp. 76–80.

[31] D. Cao, R. Ji, D. Lin, S. Li, A cross-media public sentiment analysis system for microblog, Multimedia Syst. 22 (2016) 479–486.

[32] B. Nojavanasghari, D. Gopinath, J. Koushik, T. Baltrušaitis, L.-P. Morency, Deep multimodal fusion for persuasiveness prediction, in: Proceedings of the 18th ACM International Conference on Multimodal Interaction, 2016, pp. 284–288.

[33] A. Zadeh, P.P. Liang, S. Poria, P. Vij, E. Cambria, L.-P. Morency, Multi-attention recurrent network for human communication comprehension, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 32, 2018.

[34] Q. You, L. Cao, H. Jin, J. Luo, Robust visual-textual sentiment analysis: When attention meets tree-structured recursive neural networks, in: Proceedings of the 24th ACM International Conference on Multimedia, 2016, pp. 1008–1017.

[35] F. Huang, X. Zhang, Z. Zhao, J. Xu, Z. Li, Image-text sentiment analysis via deep multimodal attentive fusion, Knowl.-Based Syst. 167 (2019) 26–37.

[36] S. Huang, T. Hui, S. Liu, G. Li, Y. Wei, J. Han, L. Liu, B. Li, Referring image segmentation via cross-modal progressive comprehension, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 10488–10497.

[37] H. Ben-Younes, R. Cadene, M. Cord, N. Thome, Mutan: Multimodal tucker fusion for visual question answering, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2612–2620.

[38] Y. Wu, P. Zhan, Y. Zhang, L. Wang, Z. Xu, Multimodal fusion with co-attention networks for fake news detection, in: Findings of the Association for Computational Linguistics, ACL-IJCNLP 2021, 2021, pp. 2560–2569.

[39] Z. Yang, J. Wang, Y. Tang, K. Chen, H. Zhao, P.H.S. Torr, Lavt: Language-aware vision transformer for referring image segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 18155–18165.

[40] J.-H. Huang, T.-W. Wu, C.-H.H. Yang, Z. Shi, I.-H. Lin, J.N. Tegnér, M. Worring, Non-local attention improves description generation for retinal images, in: 2022 IEEE/CVF Winter Conference on Applications of Computer Vision, WACV, 2022, pp. 3250–3259.

[41] J. Wang, H. Mao, H. Li, FMFN: Fine-grained multimodal fusion networks for fake news detection, Appl. Sci. (2022).

[42] X. Chen, N. Zhang, L. Li, S. Deng, C. Tan, C. Xu, F. Huang, L. Si, H. Chen, Hybrid transformer with multi-level fusion for multimodal knowledge graph completion, in: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2022.

[43] H. Mouchere, C. Viard-Gaudin, R. Zanibbi, U. Garain, ICFHR 2014 competition on recognition of on-line handwritten mathematical expressions (CROHME 2014), in: 2014 14th International Conference on Frontiers in Handwriting Recognition, IEEE, 2014, pp. 791–796.

[44] M.D. Zeiler, Adadelta: An adaptive learning rate method, 2012, arXiv preprint arXiv:1212.5701.

[45] Z. Zhang, T. He, H. Zhang, Z. Zhang, J. Xie, M. Li, Bag of freebies for training object detection neural networks, 2019, arXiv preprint arXiv:1902.04103.

[46] W. Zhao, L. Gao, CoMER: Modeling coverage fortransformer-based handwritten mathematical expression recognition, in: Proceedings of the 2022 European Conference on Computer Vision, 2022.